

ARYAN PATHANIA

Data Analyst | Data Science & Business Intelligence

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[Portfolio Website](#) | [LinkedIn](#) | [GitHub](#)

SUMMARY

Final-year B.Tech student in Artificial Intelligence & Data Science (CGC University, Mohali) with hands-on experience across the full data-to-decision pipeline — SQL/Python-based data cleaning, exploratory analysis, statistical modeling, and BI dashboarding on datasets up to 50,000+ records. Comfortable translating raw, messy data into forecasts, segment insights, and stakeholder-ready visualizations. Seeking a full-time Data Analyst or Data Scientist role.

CORE SKILLS

Languages & Tools: Python, SQL, MySQL, Microsoft Excel, Pandas, NumPy, Git/GitHub, Jupyter Notebook

Data Analysis & Statistics: Exploratory Data Analysis, Data Cleaning, Statistics & Probability, Feature Engineering, Time-Series Forecasting, Hypothesis-driven Analysis

Machine Learning: Scikit-learn, XGBoost, Regression & Classification, Model Evaluation (Precision/Recall/F1/ROC-AUC)

BI & Visualization: Tableau, Power BI (in progress), Streamlit, Matplotlib, Seaborn, Dashboard Design for Non-Technical Stakeholders

Deployment: FastAPI, REST APIs, Swagger UI, Streamlit

Foundations: DBMS, Data Structures & Algorithms, Object-Oriented Programming

PROJECTS

Sales Forecasting & BI Dashboard — Python, SQL, Tableau, Streamlit

- Designed a forecasting system over 3 years / 50,000+ transactions of historical sales data to predict revenue trends and support planning decisions.
- Cut data inconsistencies by 40% during cleaning; achieved mean absolute error within 5% of actual revenue with time-series and regression models.
- Deployed the predictive model as a live, interactive web application using Streamlit, allowing stakeholders to adjust forecast horizons and view real-time KPIs.
- Built Tableau dashboards surfacing KPIs, monthly growth trends, and region-wise performance for decision-makers.

Customer Churn Prediction API — Python, XGBoost, FastAPI, REST

- Built an end-to-end pipeline to identify churn drivers on a 7,000+ record telecom dataset, benchmarking Logistic Regression, Random Forest, and XGBoost.
- Cut dataset noise by ~30% through cleaning and feature engineering; reached 89% accuracy and 0.91 ROC-AUC with the best model.
- Deployed the trained XGBoost model as a production-ready REST API using FastAPI, exposing a /predict endpoint that returns churn probabilities and automated retention strategies.
- Documented API endpoints with Swagger UI for seamless integration and testing by engineering teams.

House Price Prediction & Market Analysis Platform — Python, Scikit-learn, Tableau

- Analyzed 10,000+ property records to model residential price drivers by location, area, and amenities.
- Improved RMSE by 18% over a baseline linear model via feature scaling and log-transformation; benchmarked Linear Regression, Decision Tree, and Gradient Boosting.
- Delivered interactive Tableau dashboards of price trends and regional comparisons for non-technical stakeholders.

Exploratory Data Analysis (EDA) Portfolio — Python, Pandas, Matplotlib, Seaborn

- Conducted end-to-end EDA on a real-world public dataset — cleaning, missing-value treatment, outlier detection, and statistical summarization.
- Uncovered non-obvious patterns via correlation heatmaps and trend analyses; published a fully reproducible, documented analysis on GitHub.

EDUCATION

B.Tech, Artificial Intelligence & Data Science — CGC University, Mohali

2024 – 2027 (Final Year) | 71% aggregate | Relevant coursework: ML, Statistics, DBMS, DSA

Senior Secondary (Class XII), CBSE — Hamirpur Public School, Himachal Pradesh (2023)

Secondary (Class X), CBSE — Hamirpur Public School, Himachal Pradesh (2021)

CERTIFICATIONS

[Deep Learning Specialization — Coursera / DeepLearning.AI \(Andrew Ng\)](#) | [AWS Academy Generative AI Foundations](#) | [End-to-End AI Project Development & Deployment Bootcamp](#)

ACHIEVEMENTS & ACTIVITIES

- Led a team through the Smart India Hackathon (SIH) internal round; project shortlisted at college level out of 30+ competing teams.
- Participated in multiple hackathons focused on data-driven problem statements, delivering working prototypes under time pressure.
- Attended workshops on Machine Learning and Data Analytics, applying learnings directly to academic projects.